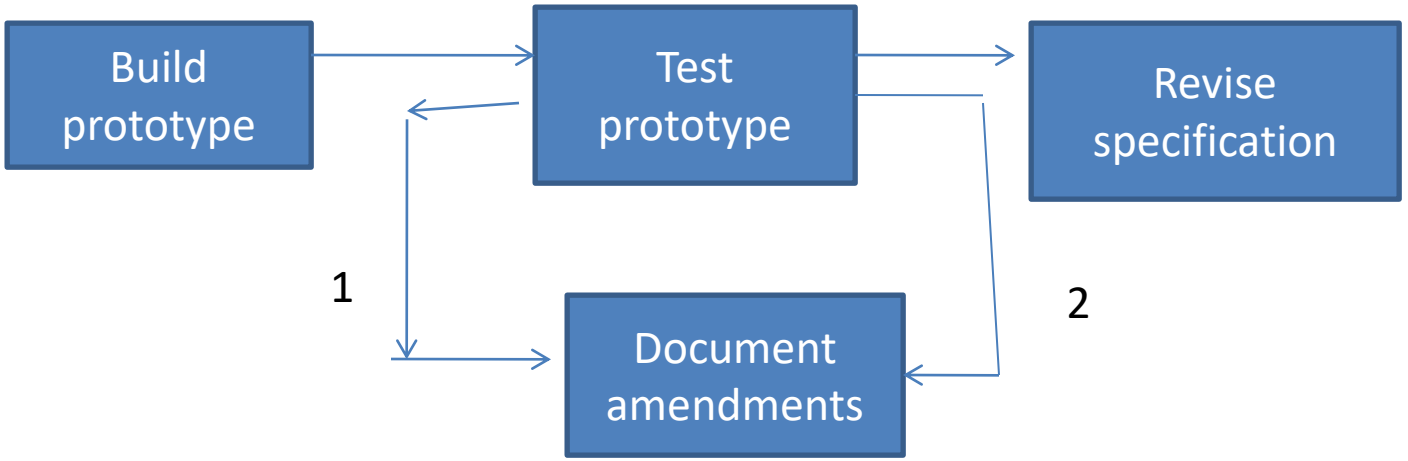
Network models: Representing lagged activities

When activities can occur in parallel with a time lag between them, the lag is represented with a duration on the linking arrow

In the Figure 1. the document amendment activity can start one day after the start of the prototype testing and will be completed two days after prototype testing is completed Figure 1.



Hammock Activities

- These have zero duration but are assumed to start at the same time as the first Hammock activity and to end at the same time as the last one
- A hammock activity is a schedule or project planning term for a grouping of tasks that "hang" between two end dates it is tied to. -Wikipedia
- They are used to represent overhead costs or other resources that will be incurred at a constant rate over the duration of a set of activities

Adding the Time Dimension

- Time at which each activity should be undertaken is to be mentioned
- The CPM (critical path method) approach is concerned with two primary objectives

1. Planning the project in such a way that it is completed as quickly as possible
2. Identifying those activities where a delay in their execution is likely to affect the overall end date of the project or later activities ‘ start dates

Adding the Time Dimension contd..

- For each activity an estimate of its duration is required
- The network is analysed by carrying out a forward pass in which the earliest dates at which activities may commence and the project be completed
- The network is then analysed by carrying out a backward pass to calculate the latest start dates for activities and to identify the critical path

Adding the Time Dimension contd..

- Generally a software is used to perform the calculations
- It is required to understand how the calculations are carried out in order to interpret the results and understand the limitations of the method

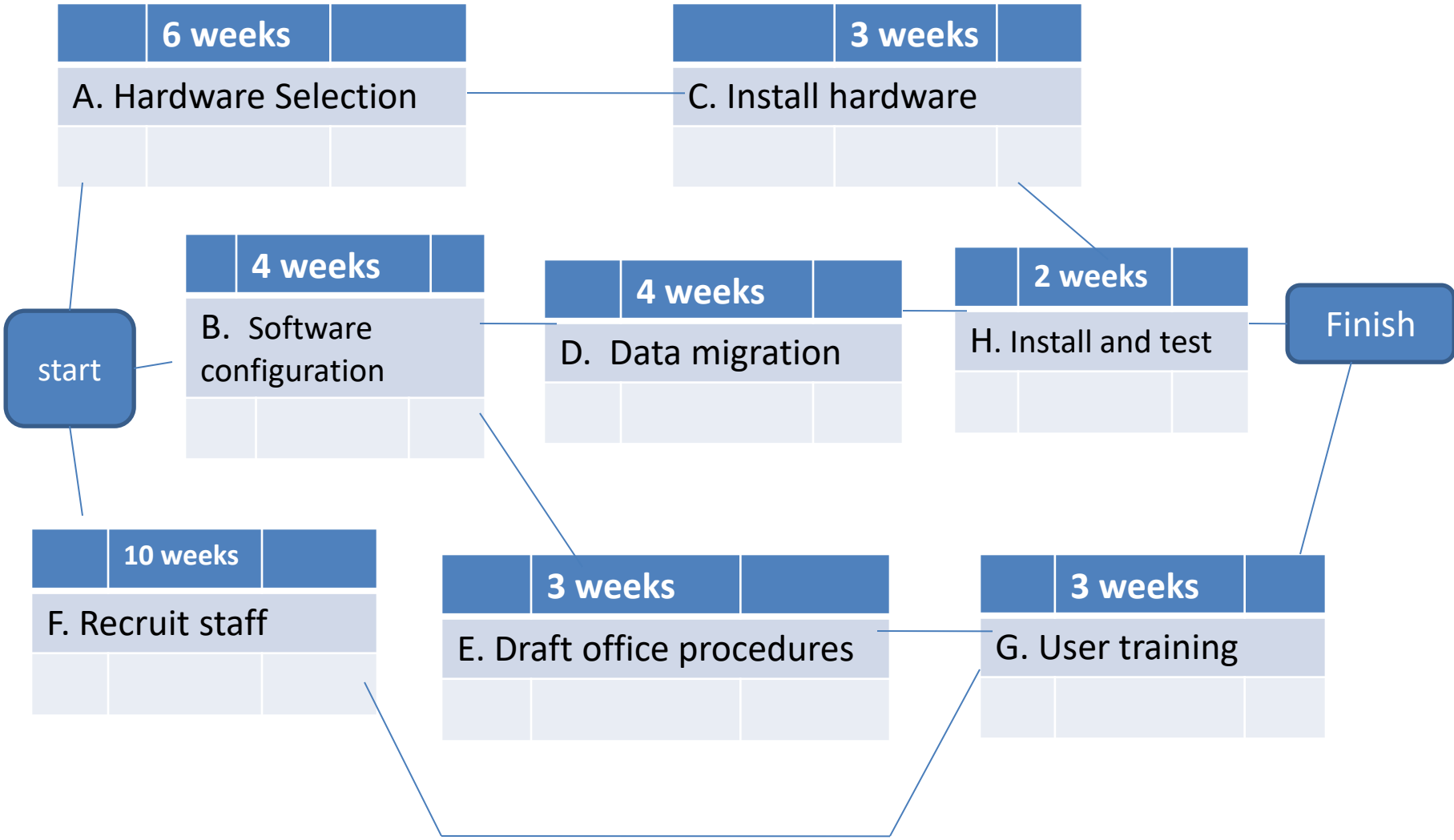
Example Project to draw an Activity Network

Problem: Consider the activities, duration in weeks and precedents in the Table 1 and draw a precedence network

Table 1.

Activity	Duration (weeks)	Precedents
A Hardware selection	6	
B System configuration	4	
C Install hardware	3	A
D Data migration	4	B
E Draft office procedures	3	B
F Recruit staff	10	
G User training	3	E, F
H Install and test system	2	C, D

Figure 2. Precedence Network



Interpretation of Figure 2.

- Start and finish indicate the beginning and end of the project
- In each box, duration is shown in the middle column of the first row; in the second row activity is shown
- The network is drawn based on the precedents; for example hardware selection, software configuration and recruit staff have no precedents and hence all the three activities have started simultaneously

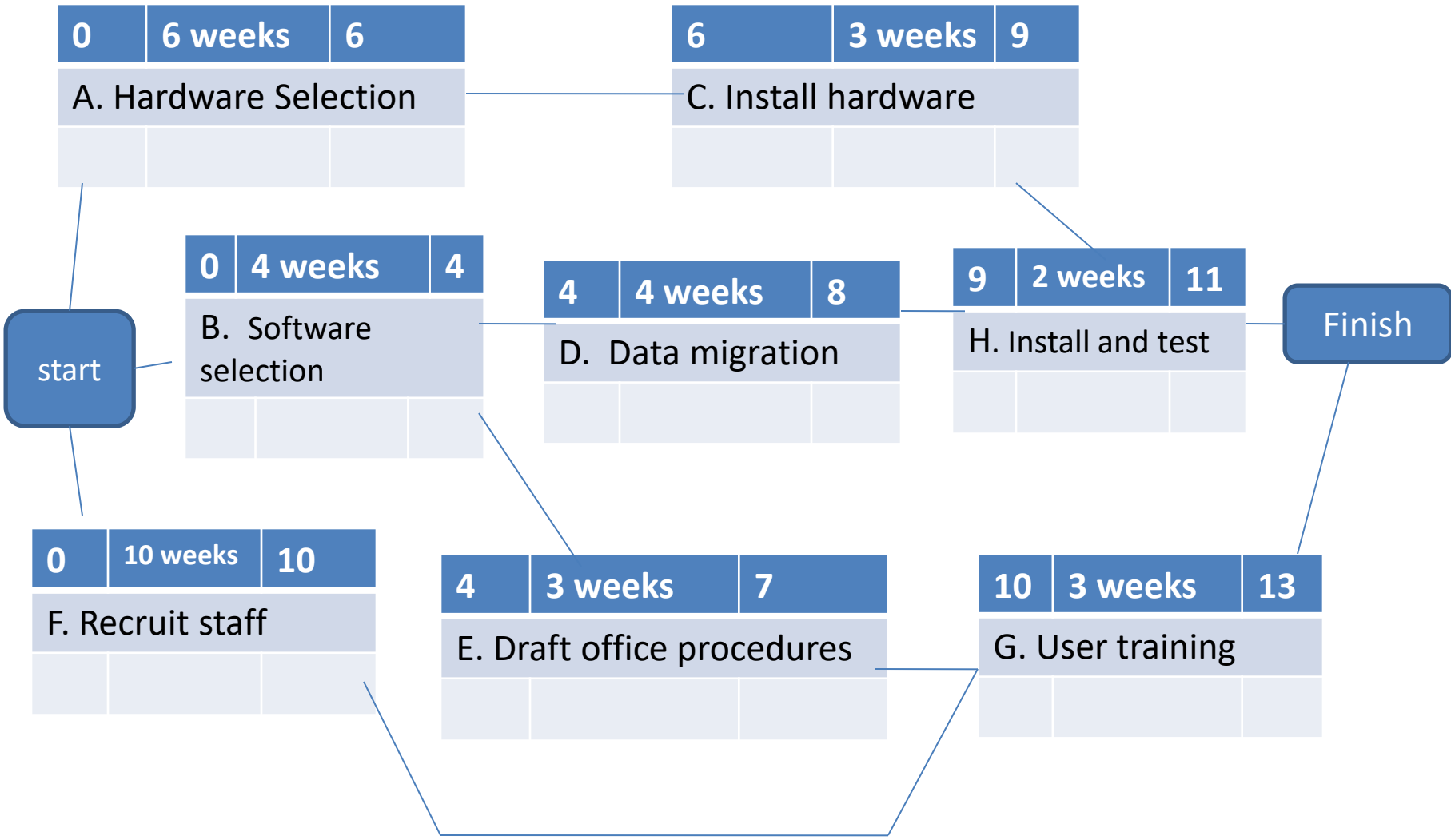
Precedents in Figure 2.

- The activity - install hardware can start after hardware selection is over as A is a precedent for C
- Similarly install and test activity can start only when software configuration and data migration are finished
- User training can start after recruit staff and draft office procedure activities are over
- By default, the project starts at the end of week 0 or at the beginning of week 1

The Forward pass

- The forward pass is carried out to calculate the earliest dates on which each activity may be started and completed
- In Figure 3. the network after forward pass is shown
- In the first row, first column project starting week and first row, third column project ending week is shown

Figure 3. Network after the Forward Pass



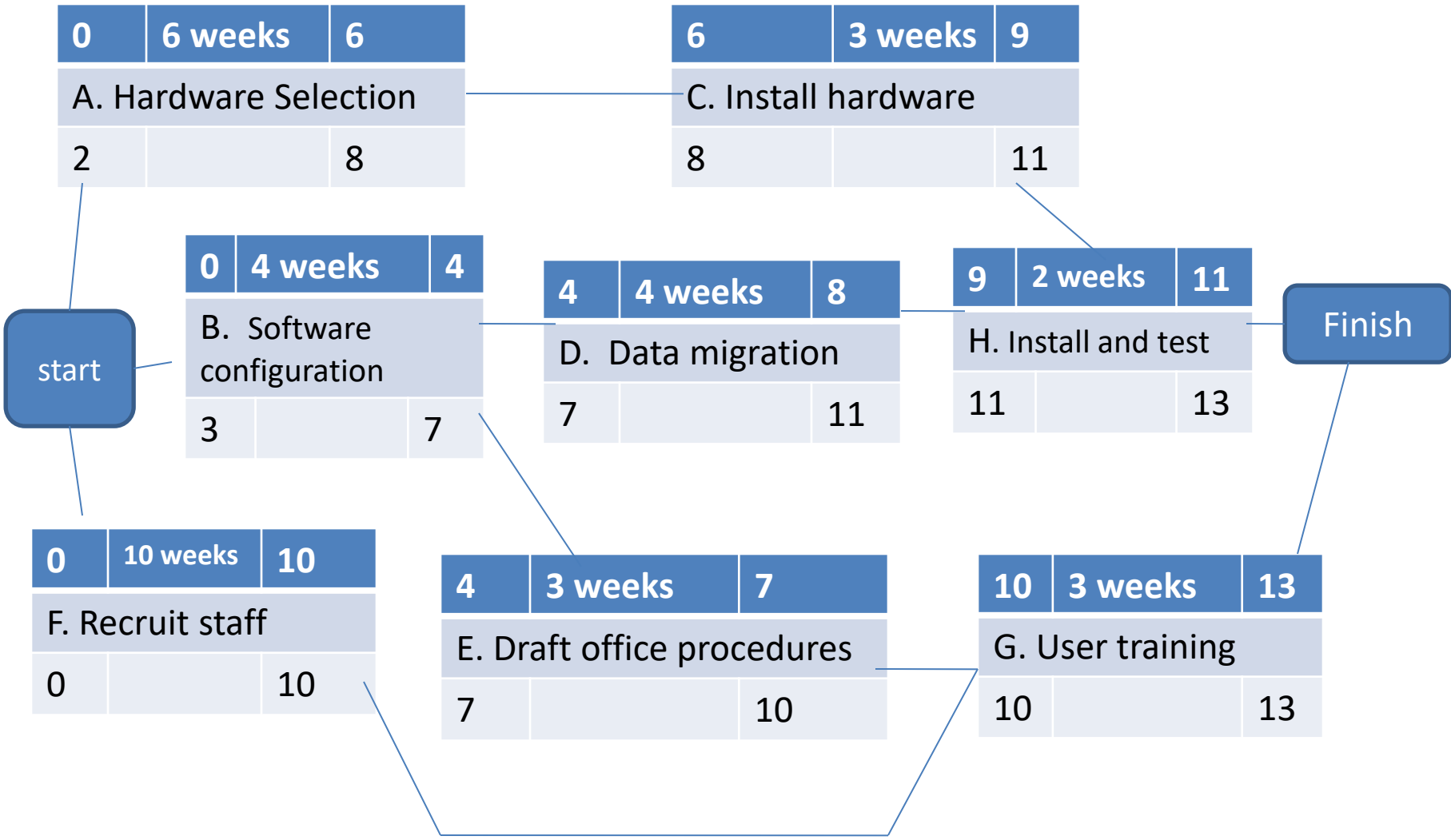
Interpretation of Figure 3.

- Hardware selection starts at week 1 and ends in week 6
- Install hardware activity starts in week 7 and ends in week 9
- Activity E i.e. Write office procedures activity takes 3 weeks and starts after activity B i.e. software configuration
- Activity B has start date 0 and end date 4 and hence Activity E has start date 4 and end date 7
- Similarly G takes 3 weeks and starts after F; hence G has start and end dates 10 and 13 respectively

The Backward pass

- The Backward pass is the second stage in the analysis of critical path network
- In this the latest date at which each activity may be started and the latest date at which each activity may be finished (without delaying the end date of the project) are calculated
- We assume that the latest finish date of the project is the same as the earliest finish date
- In the Figure 4, the network after the backward pass is shown

Figure 4. Network after the Backward Pass



Interpretation of Figure 4.

- Observe that there are numbers in the third row of each box
- The number in the first column indicates the latest start date (week) and the number in the third column indicates latest finish date (week)
- In the backward pass, we start with end activities
- For activities G and H, the latest finish date which is the same as the earliest finish date of the project is 13
- Activity H can therefore start at week 11 (= 13-2) and the latest start date for the activity G is 10 (=13-3)

Interpretation of Figure 4.

- The latest completion date for activities C and D is the latest date at which activity H must start i.e. week 11
- As the durations for C and D are 3 & 4 respectively, the earliest start dates are 8 (=11-3) & 7 (=11-4) respectively
- In a similar way, the latest start and completion dates for other activities are obtained
- The latest start date of the project is the earliest of the latest start dates of the activities A, B and F. This is week zero
- If the project does not start on time, it can not finish on time

Identifying the Critical path

- There will be at least one path through the network (i.e. one set of successive activities) that defines the duration of the project; this path is known as the critical path
- Any delay to any activity on this path will delay the completion of the project
- The critical path is the longest path through the network

Activity Float

- The difference between an activity's earliest start date and latest start date or (equally the difference between its earliest and latest finish dates) is known as the activity's float
- Activity float is a measure of how much the start or completion of an activity may be delayed
- Any activity with a float of zero is critical in the sense that any delay in carrying out the activity will delay the completion date of the project as a whole

Figure 5. Explanation

- For the activities F and G float is zero
- For F float is (0-0) or (10-10)
- For G float is (10-10) or (13-13)
- Thus F and G are critical activities
- Thus the critical path is

Start to F to G to Finish

Figure 5. The Critical Path

